



SUTHIKSHAN K

9585115142 | suthikshank@gmail.com |

<https://www.linkedin.com/in/suthikshan-k-43114528b/%7C> | <https://github.com/suthiks20>

EDUCATION

PSG Institute of Technology and Applied Research
B.E Mechanical Engineering - CGPA : 8.1

Sep 2023 - May-2027

Bharathi Matric.Hr.Sec.School
HSC-86.66%

Mar 2022 - April 2023

SKILLS SUMMARY

- **AI/ML:** TensorFlow, RAG, LangChain, CLIP, YOLOv8, LangGraph, NLP
- **Languages:** C++, Python, SQL
- **Frameworks:** FastAPI, React, Node.js, Fastify, n8n
- **Soft Skills:** Leadership(NSS), Collaboration, Communication
- **Infrastructure & Networking:** REST APIs, JSON, HTTP/HTTPS, Git, CI/CD, Basic Routing.
- **Tools:** GitHub, Postman, Vercel, Render

CORE CS FUNDAMENTALS

- System Design
- Computer Networks
- DBMS
- Operating Systems
- Object Oriented Programming System

WORK EXPERIENCE

Software Developer | Bluecode Solutions

May2026 - July 2026

Built and deployed core modules of a multi-tenant Apartment Management System (Fastify, Knex, PostgreSQL, React/TypeScript) spanning complaints, residents, finance, visitors, security, and staff — including a live AI Copilot using NVIDIA NIM (Llama 3.1) grounded in real-time database context across 10+ modules.

AI/ML Engineer | Appin Technology

Apr 2025 - May 2025

Implemented scalable data preprocessing, EDA, and machine learning pipelines in Python, developing 5+ predictive models with up to 85% accuracy and a 40% reduction in preprocessing time.

Full-Stack Developer | Arttifai Tech

Feb 2025 - Mar 2025

Built CurriSync, a full-stack curriculum management platform (Fastify, Prisma, Neon PostgreSQL, React, Tailwind) with role-based access control, JWT auth, REST APIs, and an email-driven approval workflow generating automated PDF/DOCX reports.

PROJECTS

Civic Flow - AI-Driven Civic Intelligence

Engineered a full-stack civic-issue reporting platform with an AI pipeline for image authenticity (EXIF geo/timestamp checks), CLIP + YOLOv8 relevance scoring, perceptual-hash/embedding-based duplicate clustering, and FAISS-RAG severity scoring with auto-routing. Built on Fastify, Prisma, PostgreSQL, and Socket.IO, with a React/TypeScript/Vite frontend.

Secure Biometric Authentication System

Fused FaceNet cosine-similarity face matching (synthetic Aadhaar-style records), EAR-based liveness detection, and Levenshtein-distance phishing scoring into a weighted Trust Score (0.4/0.3/0.2/0.1) driving adaptive re-authentication, on an async FastAPI/SQLAlchemy/PostgreSQL backend deployed via a React dashboard and Manifest V3 Chrome extension.

Predictive Maintenance Analytics (Bearing Prognostics)

Developed a physics-informed predictive maintenance pipeline using FrSST (ssqueezepy) for fault-frequency matching (BPFO/BPFI/BSF/FTF) on the NASA IMS Bearing Dataset, building a 1-D temporal CNN (PyTorch) to map kurtosis, entropy, and fault-match counts into a Health Index (HI) for nonlinear RUL estimation, deployed via a React/Three.js 3D dashboard.

CERTIFICATES & ACHIEVEMENTS

Ideathon 2024– Runner

Aptiv Mobility X Ideathon - Runner

Data Analytics – Accenture

Data Science-Acmegrade

Deep Learning-NPTEL